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China's centralized wastewater treatment system reaches efficiency limits in tackling water scarcity

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Centralized wastewater treatment systems (CWTSs) are increasingly expected to strengthen water security through reclaimed water reuse. Yet whether centralized expansion can simultaneously achieve cost recovery and reduce water stress remains unresolved. We assess the viability of CWTSs in China using an inventory of 10,826 wastewater treatment plants and data on wastewater generation, reuse demand and socioeconomic scenarios. Here we show that CWTS benefits are spatially bounded: only approximately 0.45 billion people live in areas where reclaimed water reuse can achieve cost recovery and reduce local water stress. Under prevailing fees of 0.9–1.1 Chinese yuan m^{-3}, cost recovery requires plants to reach break-even scales of 6,793–49,744 m^3 per day. Projected urbanization does not substantially expand viable reuse opportunities, leaving 95–99% of prospective plants in unserved areas below break-even scale. Hybrid centralized–decentralized systems can meet national targets at 25–37% lower cost. These findings support a transition from purely centralized to coordinated hybrid systems in developing regions.

Wastewater is increasingly viewed as a supplementary water source for easing pressure on conventional freshwater supplies[1–5]. Centralized wastewater treatment systems (CWTSs), which integrate wastewater collection, conveyance, treatment and reclaimed water production, have become a major route for non-potable water reuse[6,7]. Global wastewater treatment service coverage increased from 31% in 2000 to 55% in 2020[8,9], improving pollutant removal[10,11] and increasing reclaimed water supply[12,13] through advanced treatment[14,15]. This expansion of wastewater treatment services has created new opportunities to use reclaimed water in water-scarce regions[12,16]. Yet the expansion alone does not determine whether reclaimed water can effectively alleviate water scarcity.

Whether CWTSs effectively mitigate water scarcity depends on how well wastewater generation, treatment capacity, sewer access and reclaimed water demand are spatially aligned. It also depends on whether tariff revenues can cover system costs[13,17–19]. Population redistribution[17,20], industrial restructuring[21,22] and changing water-use patterns[23,24] can shift wastewater loads and reuse demand away from existing infrastructure. At the same time, the high fixed costs of treatment plants and sewer networks require sufficient utilization and stable revenue over the lifetime of CWTSs[25]. When these conditions become spatially misaligned, CWTSs may increase treated wastewater volumes without substantially improving local water security, and may remain technically functional while failing to recover costs[26–28]. Thus, the key question is not only how much wastewater can be treated but also where centralized reuse can replace enough freshwater and recover its costs.

Existing studies have provided important evidence on individual components of this system. However, few have evaluated CWTSs as

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an integrated chain spanning wastewater generation, reuse and cost recovery[18,26]. Sewer networks are commonly represented by overall coverage rates[29], although network density and layout shape both collection efficiency and capital cost[23,25,30]. Reclaimed water potential is frequently estimated from treated wastewater volumes alone[31], without accounting for local water scarcity, price competitiveness and end-use suitability[32,33]. In addition, economic assessments also rarely quantify how break-even scale and cost-recovery risk vary with wastewater tariffs, reclaimed water prices, asset lifetimes, investment costs and operating costs[34,35]. These gaps limit the ability to identify where centralized expansion can support sustainable water reuse and where decentralized or hybrid systems may be more suitable[33,36,37].

China provides a valuable national-scale case for examining these constraints. Over the past two decades, China has built the world's largest centralized wastewater treatment capacity[26,38], while socioeconomic conditions, wastewater infrastructure and water scarcity vary sharply across regions[17,24,39]. Urbanization, industrial upgrading and demographic change are further reshaping the spatial match between wastewater generation and reclaimed water use[8,10,14]. Such regional variability makes China a valuable setting for identifying where CWTSs can alleviate water scarcity economically and when centralized expansion becomes financially constrained.

We develop a process-based framework that links wastewater generation, sewer collection, treatment capacity, reclaimed water demand and tariff-based revenue (Supplementary Fig. 1). We then use this framework to assess the long-term viability of CWTSs in China. Here, we combine an inventory of 10,826 wastewater treatment plants (WWTPs) with socioeconomic, water-use, sewer-network, wastewater treatment cost and tariff data (Supplementary Fig. 2). Using this integrated dataset, we address four questions: (1) the spatial mismatches between wastewater loads, sewer collection, treatment capacity and reclaimed water demand, (2) how socioeconomic change reshapes these mismatches and the capacity of CWTSs to support local water scarcity mitigation, (3) the cost-recovery conditions for wastewater treatment and reclaimed water production and (4) the costs and benefits of continued centralized expansion versus hybrid pathways combining centralized and decentralized treatment options.

Results

Dual spatial mismatches constrain CWTS-based water scarcity mitigation

Despite extensive investment in CWTSs, their contribution to alleviating water scarcity in China remains constrained. In 2020, substantial reductions in water scarcity (change in water scarcity index (ΔWSI) >10%) were concentrated in the Northern Coastal region, with the largest relative reductions (exceeding 30%) observed in Beijing and Tianjin (Fig. 1a). Together, these areas contained approximately 88.5 million people and covered 109,820 km^2, corresponding to only 6% of China's population and 1% of its land area (Fig. 1i,j and Supplementary Fig. 3).

Even under the optimistic assumption of complete reuse of treated wastewater, the benefits of CWTSs remain geographically limited. Under complete reuse, the population exposed to substantial water scarcity alleviation would increase to 513.4 million, while the affected area would expand to 1,105,440 km^2 (Fig. 1b,i,j). Relative to current conditions in 2020, this represents a 5.8-fold increase in the affected population and an approximately 10-fold increase in the affected area. Nevertheless, nearly two-thirds of the population would remain outside areas where reclaimed water could reduce water scarcity by more than 10%.

The spatial ceiling on CWTS-based water scarcity mitigation arises primarily from the weak alignment between wastewater generation and water scarcity conditions. Only 38% of regions show spatial concordance between wastewater generation and water scarcity severity (Fig. 1c,h,k), indicating that many water scarcity hotspots do not generate sufficient wastewater to support large-scale reclaimed water production. By contrast, wastewater generation is strongly colocated with non-agricultural water use, with 87% spatial concordance (Fig. 1d,h,k). This contrast indicates that the potential demand for reclaimed water is concentrated mainly in densely populated and industrialized regions. In sparsely populated and agriculture-dominated water-scarce regions, including Northeast China and Northwest China, CWTSs have limited capacity to alleviate water scarcity, even under complete wastewater reuse.

Existing CWTS capacity matches total wastewater generation in only 55% of grid cells, reflecting a substantial spatial mismatch that limits local wastewater reuse for mitigating water scarcity. The spatial distribution of existing CWTS capacity aligns most closely with industrial output (75% spatial concordance; Fig. 1e), but shows weaker agreement with population distribution (37%) and industrial wastewater generation (49%) (Fig. 1k). Spatial mismatch between wastewater and CWTS capacity is evident in the Northern Coastal region for industrial wastewater (Fig. 1g,h), and in the Southwest region, where domestic wastewater loads driven by population hotspots (Fig. 1f) are located outside CWTS capacity clusters (Fig. 1g). These results highlight the potential to improve the spatial deployment of CWTSs for more effective water scarcity mitigation.

Changing wastewater drivers weaken CWTS utilization and reuse potential

China's mismatch between treatment capacity and wastewater generation reflects a long-term change in wastewater drivers. The temporal drivers of wastewater generation in China have shifted markedly over the past four decades, from industrial scale expansion (1980–2000) to urbanization-related changes (2000–2020) in water-use behaviour and rising per capita domestic water consumption (Fig. 2a). Concurrently, industrial pretreatment compliance rates increased from approximately 25% to 60% following the introduction of stricter regulations (Supplementary Fig. 4), notably the Pollutant Discharge Permit System in 2004 and the Water Pollution Prevention and Control Law in 2008. These regulations enabled nearly 20% of wastewater to be discharged directly after meeting discharge standards in 2020 (Fig. 2b). Consequently, despite continued industrial output growth of 1.7–1.8 trillion Chinese yuan (CNY) yr^{-1} during 1980–2020, industrial wastewater discharges declined sharply, from an increase of 1.2 km^3 yr^{-1} in 1980 to a decrease of 1.1 km^3 yr^{-1} in 2020 (Fig. 2c). Domestic wastewater, especially from urban areas, has therefore become the main source requiring centralized treatment.

By 2020, total WWTP design capacity had expanded to 98.04 km^3 yr^{-1}, nearly matching total annual point-source wastewater discharge, yet actual treatment remained below two-thirds of installed capacity (Fig. 2c). Approximately one-quarter of urban WWTPs and nearly half of rural WWTPs operated under suboptimal load conditions in 2020 (Fig. 2d). Nearly 15% of sewage was treated by WWTPs operating above their design capacity, whereas about 7% was treated by WWTPs running at low load (Fig. 2d). In rural areas, population outmigration led to low-load operation at nearly half of WWTPs, while approximately 63% of rural sewage was discharged directly into the environment without adequate treatment (Fig. 2e). The main challenge was therefore not simply insufficient capacity, but a mismatch among plant capacity, sewer access, population change and wastewater sources.

The shift from industrial to domestic wastewater has increased the risk of mismatch between wastewater supply and demand for reclaimed water. In total, potentially substitutable industrial and domestic water use exceeded wastewater generation during 1980–2020 (Fig. 2f). However, national totals masked important local and sectoral constraints. In 2020, domestic wastewater accounted for 84% of total wastewater generation, whereas reclaimable domestic demand was limited mainly to non-potable uses and represented only 20–33% of non-agricultural water demand (Fig. 2g). By contrast, the energy-production sector contributed around half of total water demand eligible for reclaimed

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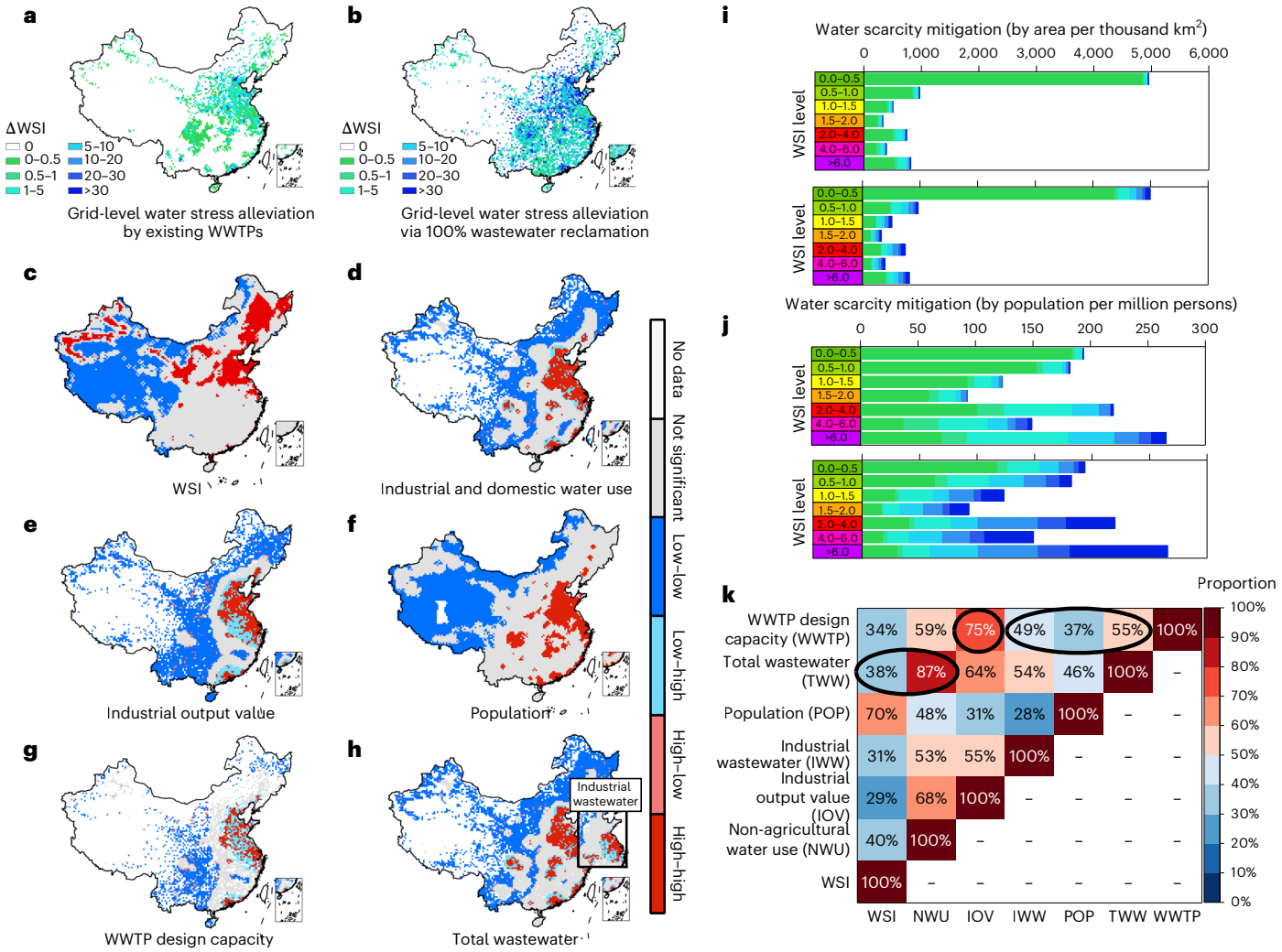


Fig. 1 | Spatial limits to CWTS-driven water scarcity mitigation. a,**b**, Grid-level reductions in water scarcity attributable to CWTSs under current wastewater reuse (**a**) and a hypothetical 100% reuse scenario (**b**). **c**–**h**, Spatial clusters of wastewater-related factors, including the WSI (**c**), industrial and domestic water use (**d**), industrial output value (**e**), population (**f**), WWTP design capacity (**g**) and total wastewater generation (**h**), with industrial wastewater shown in the inset. **i**,**j**, Land area (**i**) and population (**j**) benefiting from CWTS-mediated water scarcity mitigation across WSI levels under current and full-reuse conditions. **k**, Pairwise spatial matching among wastewater-related factors. Basemap data in **a**–**h** from the National Platform for Common GeoSpatial Information Services (https://www.tianditu.gov.cn) (approval number: GS(2024)0650).

water substitution because cooling water accounted for about 95% of its withdrawals (Fig. 2g and Supplementary Fig. 5). Thus, the most valuable reuse opportunities remained concentrated in specific industrial sectors and densely populated cities, rather than increasing together with domestic wastewater.

Future regional socioeconomic change does not expand the effective reach of CWTSs

Future population redistribution, urbanization and economic growth are projected to have only limited effects on the areas where CWTSs can both recover costs and reduce water scarcity (Fig. 3a). Although wastewater collection and reclaimed water reuse are projected to increase in some regions, these gains remain concentrated in locations that already have dense populations, industrial activities and mature infrastructure. Future socioeconomic change therefore does not eliminate the spatial constraints identified above, because it neither redistributes wastewater towards regions with surplus treatment capacity nor creates sufficient economically viable reclaimed water demand in many water-scarce regions.

Although continued urbanization is projected to increase the population served by mature wastewater collection networks by 0.1–0.2 billion by 2050 (Fig. 3c), this improvement in collection access does not translate into a more balanced distribution of WWTP design capacity. We identify a robust S-shaped relationship between collection-network density and wastewater collection rate, revealing distinct threshold regimes (Fig. 3b, Supplementary Note 4 and Supplementary Fig. 6). Collection-network densities below 5.2 km km^{-2} correspond to underdeveloped systems, whereas those exceeding 9.7 km km^{-2} indicate mature systems with diminishing marginal returns. Achieving near-complete wastewater collection requires densities approaching 18 km km^{-2}, comparable to those observed in high-income countries[19]. Because population inflows are concentrated in already urbanized coastal regions with mature networks (Supplementary Fig. 7), additional wastewater generation primarily increases collection loads in areas where treatment demands already approach or exceed available design capacity. Consequently, capacity deficits persist in the Eastern and Southern Coastal regions, while widespread surplus capacity remains in less densely populated northern regions (Fig. 3a and Supplementary Fig. 8).

Future socioeconomic growth is also unlikely to alter the spatial concentration of stable, higher-tariff reclaimed water demand. Under prevailing reclaimed water tariffs, achieving cost recovery with 90–95%

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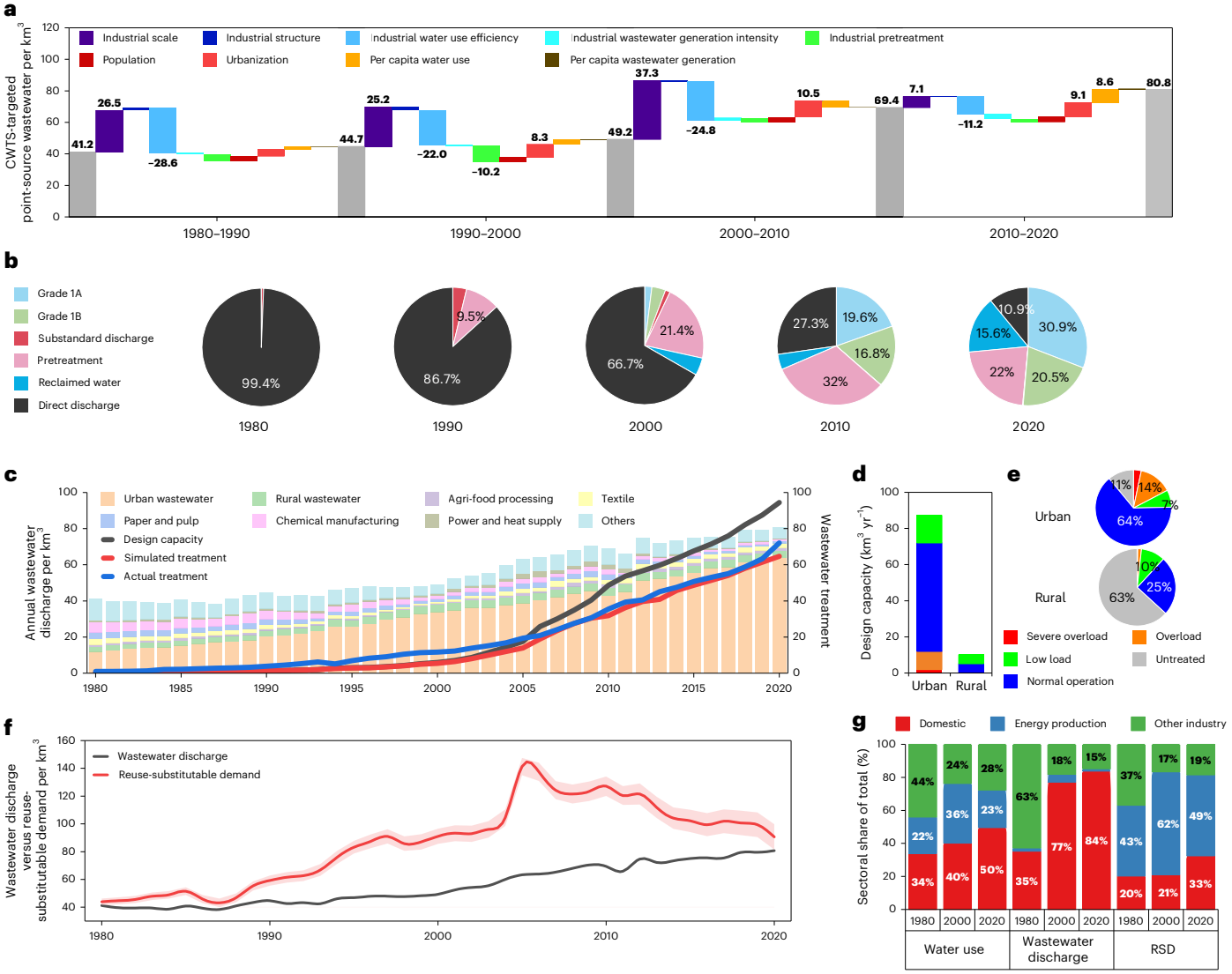


Fig. 2 | Process-based wastewater simulation chain and temporal changes from 1980 to 2020. a, The contribution of nine socioeconomic drivers to the change in the point-source wastewater discharge during 1980–2020. For detailed information, please refer to Supplementary Table 1 and Supplementary Fig. 4. **b**, The destinations of wastewater from 1980 to 2020. For detailed information, please refer to Supplementary Table 2. **c**, Changes in point-source wastewater discharge and CWTS design capacity from 1980 to 2020. For detailed information, please refer to Supplementary Table 3. **d**, Evaluation of the operation status of urban and rural WWTPs in 2020 (normal operation: ≥60 and <100%; low load: <60%; overload: ≥100 and <120%; and severe overload: ≥120%). Technical thresholds are defined by Chinese technical guidelines[80,81]. **e**, Urban (top) and rural (bottom) wastewater treatment status in 2020. **f**, Temporal change between water demand eligible for reclaimed water substitution and wastewater discharge from 1980 to 2020. In **f**, the central lines indicate median (P_{50}) estimates, and the shaded band indicates the 5th–95th percentile range across 1,000 Monte Carlo realizations. **g**, Sectoral shifts in water use, wastewater generation and water demand eligible for reclaimed water substitution in 1980, 2000 and 2020. RSD, reuse-substitutable demand.

confidence would require allocating all produced reclaimed water to industrial cooling and non-potable uses (Fig. 3e and Supplementary Fig. 9). These profitable and stable uses remain concentrated in industrialized and densely populated regions, including the North China Plain, the middle and lower Yangtze River basin, the southeastern coast and the Pearl River Delta (Fig. 3d and Supplementary Fig. 10). Future socioeconomic growth therefore increases profitable reuse potential in water-scarce regions by only 5–10 km^3 by 2050, without materially changing its spatial distribution (Fig. 3f). In particular, future development does not generate sufficient stable reuse demand in sparsely populated and agriculture-dominated water-scarce regions.

Under the joint constraints of water scarcity mitigation, cost recovery and the utilization target, 56–71% of national wastewater generation can be cost-effectively reused (Fig. 3d). Reusing this share can alleviate water scarcity for 420–500 million people. Relaxing the cost-recovery constraint expands the spatial extent of potential reuse, mainly in the middle reaches of the Yangtze River basin, and adds 290–330 million potential beneficiaries. However, the financial viability of these additional gains is more uncertain. Conversely, prioritizing cost recovery alone allows about 78% of wastewater to be converted into higher-tariff reclaimed water uses, but this does not necessarily coincide with regions where reuse would most effectively alleviate water scarcity. Northeast and West China meet neither the cost-recovery criterion nor the water-scarcity-mitigation criterion. In these regions, fragmented local demand, limited wastewater availability and irrigation-dominated water scarcity constrain both reliable reclaimed water revenues and substantial water scarcity alleviation.

Economic constraints limit the effective expansion of CWTSs

Under prevailing wastewater tariffs and cost conditions, extending CWTS coverage to unserved areas would generally be financially infeasible. Across different cost-revenue scenarios, 60–99% of prospective

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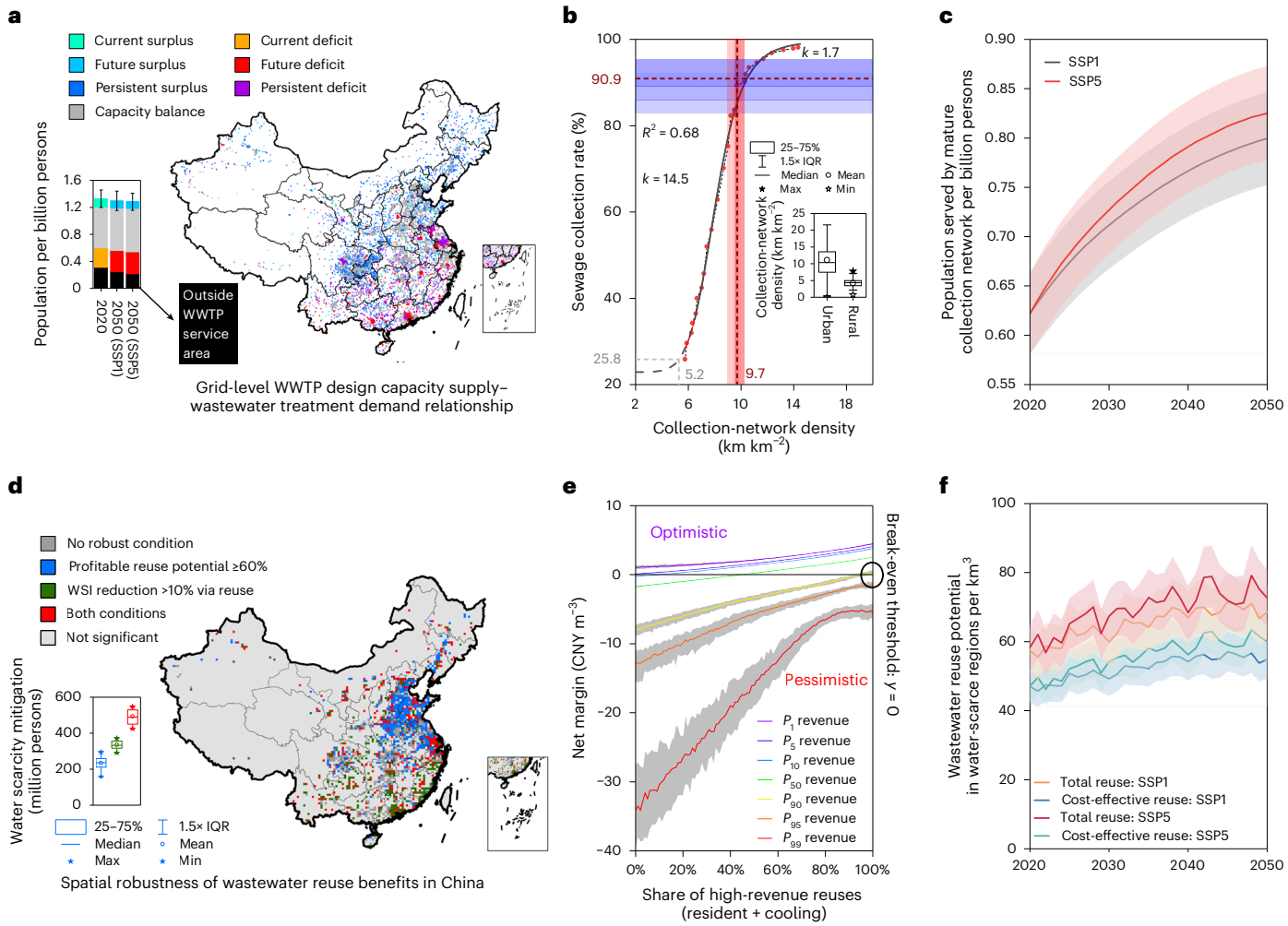


Fig. 3 | CWTS-related infrastructure and demand constraints on wastewater collection, treatment capacity and reuse benefits in China. a, Spatial mismatch between WWTP design capacity and wastewater treatment demand. **b**, Nonlinear relationship between collection-network density and wastewater collection rate. **c**, Population served by mature collection networks under SSP scenarios. **d**, Spatial robustness of wastewater reuse benefits for water scarcity mitigation. A 60% reclaimed water utilization rate is a national policy target set by the Chinese government for 2035 (ref. 41). **e**, Cost-recovery probability constrained by cost-effective reclaimed water uses. **f**, Future wastewater reuse potential in water-scarce regions. In the inset of **a**, bars indicate means, and error bars indicate standard deviations based on 1,000 Monte Carlo simulations for each category. In the insets of **b**, **c**, **e** and **f**, the lines indicate median (P_{50}) estimates, and shaded areas represent the 90% uncertainty intervals of the estimated mitigation costs and benefits, derived from 1,000 Monte Carlo simulations. For the inset box plots in **b** and **d**, centre lines indicate medians, boxes span the 25th–75th percentiles and whiskers extend to the most extreme observations within 1.5× the interquartile range (IQR). Sample sizes are 341 and 31 urban and rural observations in **b**, respectively, and 1,000 Monte Carlo realizations per category in **d**. Basemap data in **a** and **b** from the National Platform for Common GeoSpatial Information Services (https://www.tianditu.gov.cn) (approval number: GS(2024)0650).

WWTP design capacity and 95–99% of prospective WWTPs in unserved areas fall below the cost-recovery threshold (Fig. 4f). These financial risks exceed those associated with existing CWTS assets. Although the economic performance of existing systems is also unfavourable, only 8–39% of their design capacity and 70–93% of their WWTPs face a risk of financial unsustainability (Fig. 4f). This contrast highlights the structural financial disadvantage of CWTS expansion into unserved areas.

This disadvantage arises from two compounding cost penalties in CWTS investment, namely capital costs of WWTPs and sewer networks. As WWTP design capacity increases, the unit capital cost of WWTPs and its sensitivity to net cost decline markedly (Fig. 4a,b). Dispersed settlements rapidly increase both the cost and sensitivity of connecting wastewater sources to sewer networks (Fig. 4b,e). These contrasts are particularly pronounced along the urban–rural gradient. In urban areas, treatment-capacity expansion is the dominant driver of variation in fixed-asset investment, accounting for 64.8%, whereas pipe construction accounts for only 4.9% (Fig. 4d). Rural areas show the opposite pattern: pipe construction becomes the primary cost driver, accounting for 58.7%, followed by treatment-capacity expansion at 27.5% (Fig. 4d).

The spatial characteristics of unserved areas formed a particularly unfavourable combination of low wastewater volumes and high connection-pipeline requirements. Within a standardized 25-km^2 window (approximately equivalent to the minimum catchment area of a CWTS; Supplementary Fig. 11), daily wastewater generation averaged 8,802 m^3 per day in served urban areas but only 1,955 m^3 per day in unserved urban areas. Corresponding volumes decreased further to 360 m^3 per day and 181 m^3 per day in served rural and unserved rural areas, respectively (Fig. 4e). Meanwhile, required connection-pipeline density increased from 1.1 km km^{-2} in served urban areas to 3.9 km km^{-2} in unserved urban areas, reaching 5.1 km km^{-2} in unserved rural areas (Fig. 4e). Unserved areas therefore lacked sufficient wastewater volumes to spread the fixed costs of treatment infrastructure,

while requiring longer connecting networks to aggregate dispersed wastewater sources.

These spatial characteristics and cost differences ultimately defined a strict economic feasibility boundary on centralized CWTS expansion. Below a wastewater tariff of 0.8 CNY m^{-3}, cost recovery was effectively unattainable across the range of plant sizes considered (Fig. 4a,c). Even within the tariff range of 0.9–1.1 CNY m^{-3} currently applied in many Chinese cities, cost recovery was possible only under the most favourable cost conditions and for CWTSs treating approximately 6,793–49,744 m^3 per day (Fig. 4a,c). Sensitivity analysis showed that the precise location of this threshold is mainly determined by sewer-network and WWTP design lifetimes, together with scale-dependent capital costs. Nevertheless, sensitivity analyses consistently indicated that the centralized expansion model has reached its economic feasibility boundary under prevailing tariff conditions.

Hybrid systems overcome the economic limits of centralized expansion

The economic boundary identified above does not render China's wastewater treatment and reclaimed water utilization targets unattainable. Instead, it identifies where continued centralized expansion should be replaced by decentralized alternatives. The hybrid scheme (HS; decentralized-dominant with centralized augmentation) assigns dispersed, low-volume wastewater to decentralized facilities and reserves centralized facilities for areas with concentrated wastewater generation. It meets the 2035 targets[40,41] and maintains them through 2050 at substantially lower cost than the only-CWTS scheme (OC) (Fig. 5a). Under OC, closing treatment and reuse gaps required an additional investment of 4.0–5.0 trillion CNY across Shared Socioeconomic Pathway (SSP)1 and SSP5, including 1.9–2.2 trillion CNY in unserved areas. By contrast, the HS reduced total costs by 25–37% relative to continued centralized expansion and lowered net unit treatment costs by 0.30–0.96 CNY m^{-3} by 2050.

These savings arose because the HS avoided long collection sewers and small-scale WWTPs in sparsely populated rural areas, where unit costs rose sharply. It shifted part of rural wastewater treatment expansion from conventional sewer extensions to decentralized facilities, notably subsurface-flow-constructed wetlands (119.7–120.4 billion CNY), surface-flow-constructed wetlands (29.7–30.5 billion CNY) and stabilization ponds (78.4–79.6 billion CNY). The HS avoided approximately 1 trillion CNY that would otherwise be required for new centralized WWTPs and sewer networks. The HS incurred approximately 3.5 billion CNY yr^{-1} in additional reconstruction costs due to the shorter lifespans of decentralized systems, and experienced an annual revenue loss of approximately 7.0 billion CNY as future urbanization shifted wastewater flows away from some initially deployed rural facilities (Fig. 5b). Nevertheless, these additional costs were substantially smaller than the wastewater treatment and sewer-network expenditures avoided by replacing centralized expansion in low-density areas.

Investment timing further affected the economic performance of HS. Compared with the investment-capped (IC) strategy, which applies a fixed annual budget, the goal-oriented (GO) strategy reduced net unit treatment costs by approximately 0.10–0.30 CNY m^{-3} across scenarios (Fig. 5a). Under the IC strategy, early investment generated an initial reclaimed water revenue of approximately 5.63 billion CNY yr^{-1}, but subsequent changes in population distribution and industrial water demand reduced annual reclaimed water revenue by approximately 16.98 billion CNY yr^{-1} by 2050 (Fig. 5b). The GO strategy avoided part of this loss by delaying or redirecting investment until wastewater generation and profitable reuse demand were more firmly established. This flexibility required higher peak annual investment, approximately 3.7–4.5 times that under the IC strategy, to close treatment and reuse gaps by 2035 (Supplementary Fig. 16). Prioritizing net economic benefits did not materially reduce water scarcity mitigation: strategies optimized for either net benefits or substantial water scarcity alleviation (ΔWSI >10%) produced similar unit costs and substantially alleviated water scarcity for approximately 460 million people (Supplementary Figs. 17 and 18).

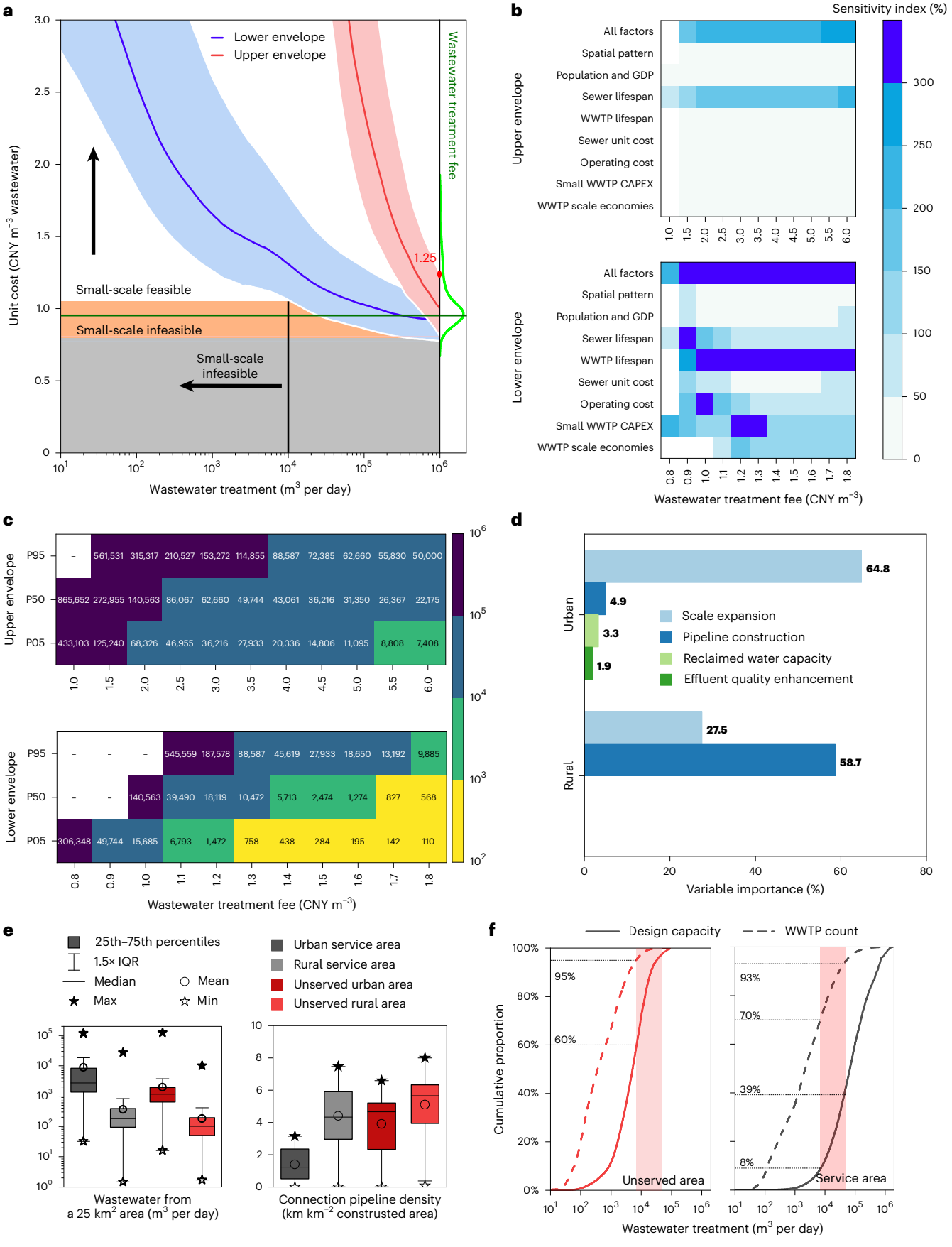
The HS also improved the financial robustness of the national wastewater treatment system. Relative to the current configuration, it reduced the share of WWTPs that were unlikely to recover costs by 7–22 percentage points, while the OC increased this share by approximately 1–7 percentage points (Fig. 5c). Centralized adjustment alone could reduce cost-recovery risk for only around 2% of design capacity by connecting low-load systems to nearby WWTPs, while approximately 6% of capacity remained economically difficult to improve through further centralized consolidation. By reallocating dispersed demand to decentralized facilities, the HS reduced the share of treatment capacity below the break-even scale from approximately 8% to 4% (Fig. 5c). Its advantage therefore extended beyond lower aggregate investment to reduced exposure of the wastewater treatment system to plant sizes that were unlikely to recover costs. The cost advantage of HS remained robust across uncertainty scenarios: shortening asset lifetimes from 40 to 20 years increased net unit treatment costs by approximately 0.30–0.35 CNY m^{-3}, and variation in fixed-asset investment and revenue changed costs by up to approximately 0.20–0.30 CNY m^{-3}, but HS remained less costly than OC across all tested parameter combinations (Fig. 5d and Supplementary Fig. 19).

Hybrid wastewater system deployment followed a broadly similar temporal sequence across China. During 2021–2022, almost all regions experienced a surge in investment in treatment-capacity expansion to address immediate treatment deficits (Fig. 6a). Between 2023 and 2025, the share of investment allocated to decentralized systems increased, particularly for subsurface-flow-constructed wetlands, to avoid costly sewer extensions in dispersed settlements. From 2026 to 2035, sewer-related investment gradually increased in areas with sustained wastewater treatment demand. This later expansion occurred only after local treatment capacity and service demand had become sufficiently established, reducing the risk of excessive upfront investment in sewer networks.

The allocation and duration of investment differed substantially between northern and southern China (Fig. 6a). Northern regions allocated a larger and more sustained share of investment to decentralized facilities than southern regions: throughout 2021–2027, more than 20% of annual investment in northern China was directed towards decentralized wastewater treatment facilities. In southern China, investment was more strongly concentrated in treatment-capacity expansion. Except for a short period of concentrated decentralized investment during 2023–2025, more than 30% of annual investment in most years was

Fig. 4 | Scale-dependent cost constraints on CWTS investment. a, Unit-cost envelopes and feasibility thresholds for wastewater treatment across plant scales. The lower and upper break-even cost envelopes were derived from two-level Monte Carlo simulations (Supplementary Note 7). The solid curves show the median envelopes, and the shaded bands indicate their 5th–95th percentile ranges. At a given treatment scale, a wastewater treatment fee below the lower envelope cannot recover costs, whereas a fee above the upper envelope guarantees cost recovery. For detailed information, please refer to Supplementary Fig. 12. **b**,**c**, Sensitivity of cost-envelope estimates (**b**) and treatment-scale thresholds under alternative wastewater treatment fees (**c**). **d**, The importance of each component of CWTSs to fixed-asset investment. **e**, Differences in wastewater generation and connection-pipeline density between urban and rural service areas and currently unserved areas. In **e**, sample sizes are 21,038, 30,100, 9,716 and 35,419 25-km^2 spatial analysis units for urban service, rural service, unserved urban and unserved rural areas, respectively. For detailed information, please refer to Supplementary Figs. 13–15. **f**, WWTPs below the economic threshold in service versus unserved areas: plant counts and design capacities. GDP, gross domestic product; IQR, interquartile range. CAPEX, capital expenditure.

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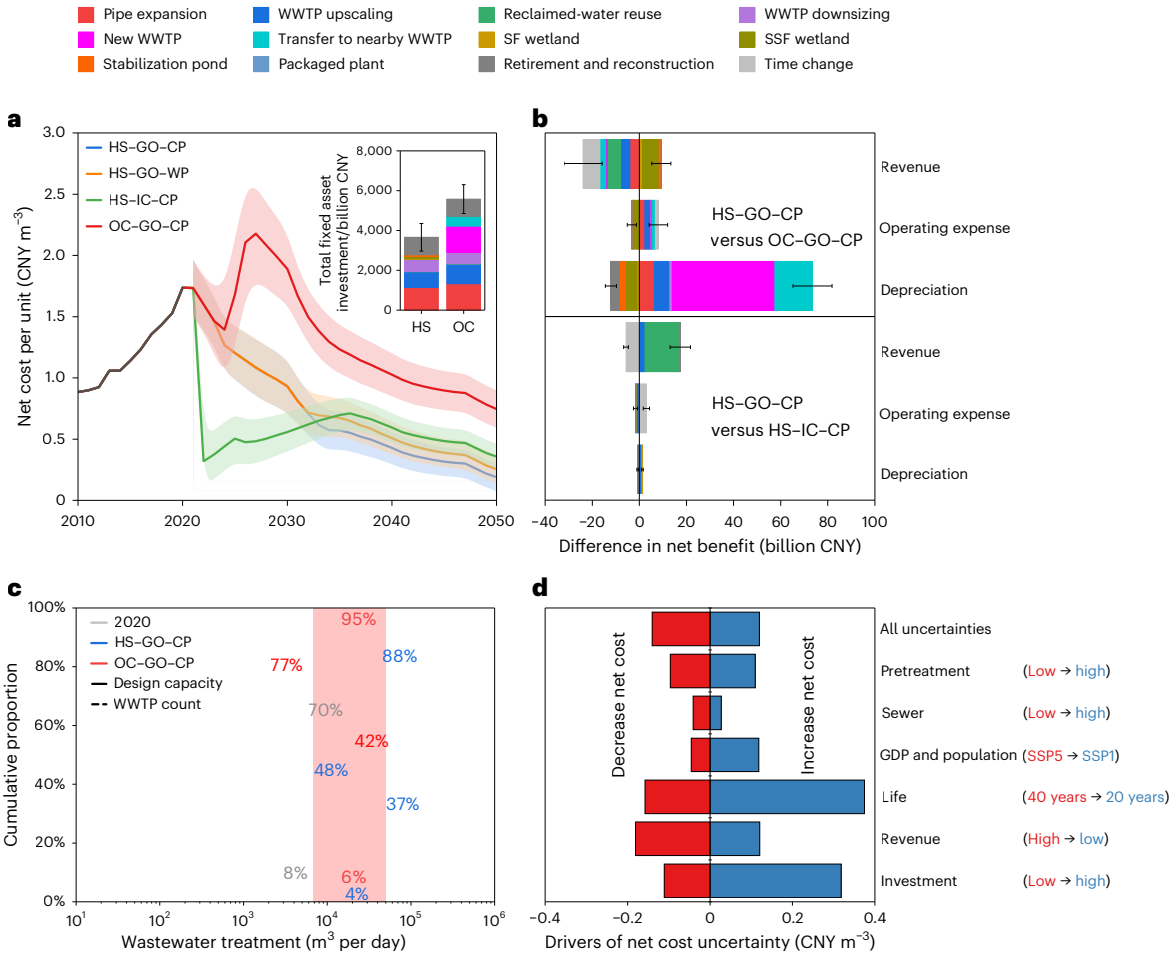


Fig. 5 | Economic trade-offs and structural adjustment of China's CWTS transition pathways. a, Unit net cost and investment composition under HS and OC pathways. **b**, Net-benefit component differences. **c**, Shifts in WWTP scale structure relative to economic thresholds. **d**, Drivers of net-cost uncertainty. The labels indicate pathway–strategy–priority combinations: HS, OC, GO, IC, cost-prioritized (CP) and water-scarcity-prioritized (WP). In **a**, solid lines indicate median (P_{50}) estimates, and shaded bands indicate the 5th–95th percentile ranges across 1,000 Monte Carlo realizations for each pathway–strategy–priority combination. In **a** and **b**, the bars indicate means, and error bars indicate standard deviations across 1,000 Monte Carlo realizations. In **d**, the bars show the 5th–95th percentile range of changes in net unit cost when each uncertain parameter is varied independently, with negative and positive values indicating decreases and increases relative to the baseline, respectively. SF, surface-flow constructed wetland; SSF, subsurface-flow constructed wetland.

allocated to expanding centralized treatment capacity. Technology selection also varied among regions. Subsurface-flow-constructed wetlands were generally the most cost-effective decentralized option in all regions except Northwest China.

Spatial investment patterns did not align with where the HS delivered its greatest economic benefits relative to continued centralized expansion. Approximately 70–90% of total investment was concentrated in southern regions with relatively low water scarcity severity and in densely populated northern coastal regions, where high wastewater volumes, stable reclaimed water demand and stronger revenue-recovery conditions supported capital-intensive wastewater infrastructure (Fig. 6b). By contrast, the largest economic advantage of the HS occurred in northern China, where annual net-benefit gains exceeded those under the OC by more than 6 billion CNY (Fig. 6a). This divergence highlights a key planning implication of hybrid systems: investment should follow concentrated service demand, whereas decentralized deployment delivers its largest avoided-cost benefits in low-density areas.

Discussion

Our analysis shows that China's centralized wastewater treatment has reached a hard economic boundary in alleviating water scarcity. Cost estimates along the treatment lifecycle are subject to propagated uncertainties; we therefore estimated empirical distributions for the major cost parameters and propagated both distributional uncertainty and input variability through a two-level Monte Carlo framework. By combining these simulations with restricted cubic spline cost frontiers, we derived a range of minimum viable WWTP sizes rather than a single fitted cost curve.

The estimated threshold range of 6,793–49,744 m^3 per day defines the economic boundary of centralized treatment (Fig. 4a). Below this range, prevailing tariff structures are insufficient to satisfy the minimum cost-recovery requirement without public support, profitable reuse demand or alternative service configurations. Under the scale thresholds identified here and current wastewater treatment tariffs (0.9–1.1 CNY m^{-3}) in many Chinese cities, most small and dispersed systems fall below the cost-recovery threshold. Nearly 95% of WWTPs required in unserved areas would not meet the cost-recovery criterion, and even among existing operational plants, approximately 70–93% remain exposed to economic risk. This implies that extending centralized infrastructure into low-density or weak-demand areas requires explicit public financing[42], cross-subsidies[43] or alternative decentralized configurations[32,33,37], rather than relying on larger centralized systems to become financially self-supporting.

The estimated threshold is comparable to the scales reported in previous studies of wastewater treatment economies of scale. A

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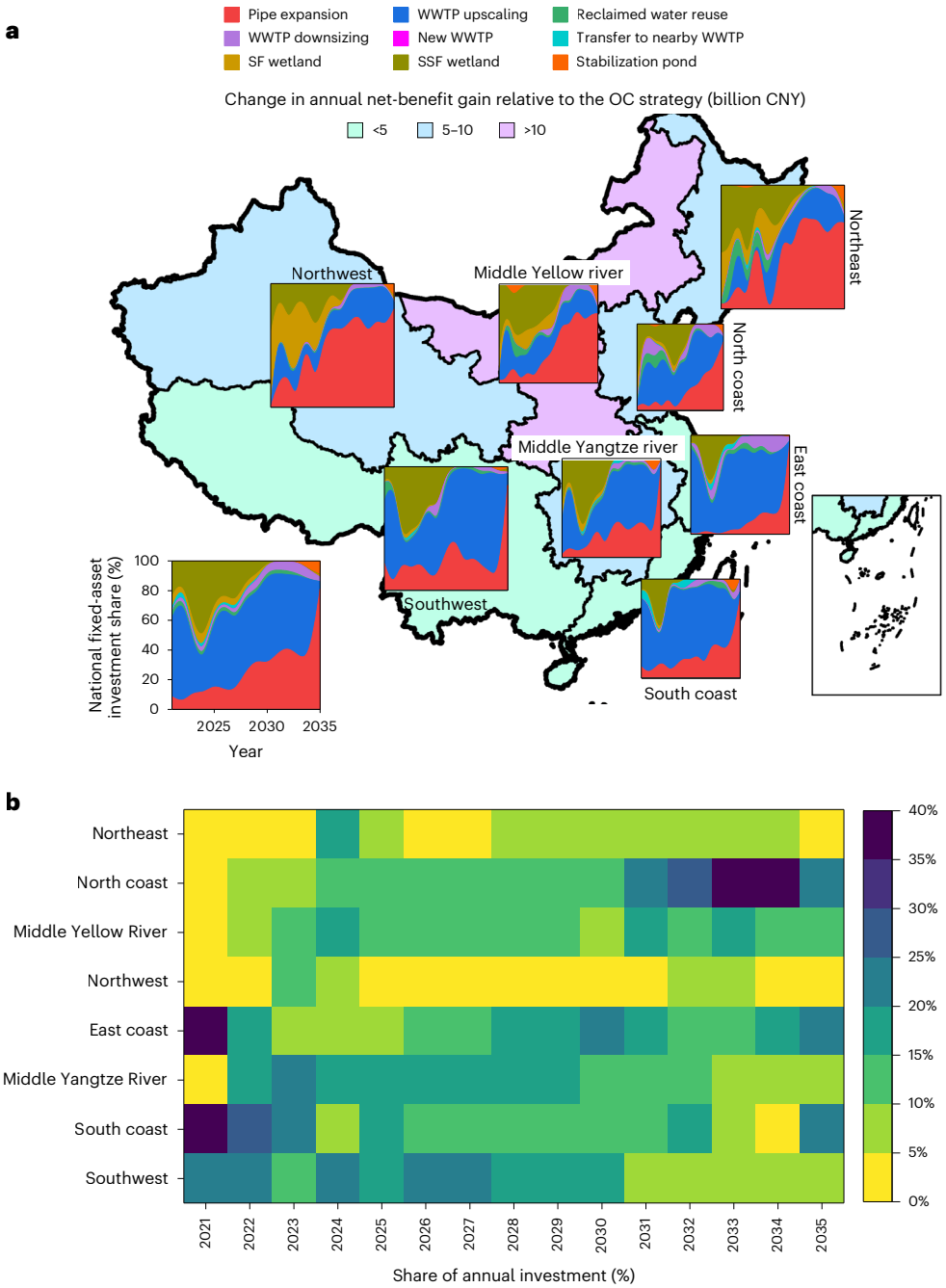


Fig. 6 | Regional investment allocation and annual net-benefit gains in China's CWTS transition. a, Regional changes in annual net-benefit gains relative to the OC strategy and regional investment composition over time. **b**, Annual investment shares by region. Basemap data in **a** from the National Platform for Common GeoSpatial Information Services (https://www.tianditu.gov.cn) (approval number: GS(2024)0650).

techno-economic analysis of wastewater reclamation in Hong Kong estimated a break-even water-reclamation scale of approximately 63,769 m^3 per day (ref. 44), higher than our range, which is plausible given Hong Kong's substantially higher labour and construction costs relative to mainland China. Evidence from Spain suggests that around 10,000 population equivalent (roughly 1,000–4,000 m^3 per day depending on assumed wastewater generation) represents a lower bound for improved efficiency. Achieving very low unit costs comparable to China's current wastewater tariffs requires substantially larger plants, around 50,000 population equivalent (approximately 5,000–20,000 m^3 per day)[45]. Evidence from Italy similarly indicates a minimum efficient scale of about 41,000 m^3 per day, serving roughly 100,000 people[46]. Although these studies differ in institutional setting, accounting scope and treatment objectives, they show a consistent pattern of economies of scale: facilities below several thousand cubic metres per day are more likely to face weak cost recovery, whereas larger systems are better able to spread fixed capital and operational costs.

Our results suggest that centralized wastewater reuse cannot be expanded uniformly as a national solution to water scarcity (Fig. 3d). Reclaimed water expansion can simultaneously recover costs and meaningfully reduce local water scarcity only where four conditions coincide: sufficient wastewater supply[47,48], concentrated non-potable demand[12,49], meaningful local water stress[50,51] and wastewater treatment scales large enough to support cost recovery (Fig. 4a). Areas satisfying all conditions are most pronounced in densely populated and industrialized coastal regions[11] and in parts of the middle Yangtze River basin[52] (Fig. 1e,f), where non-potable municipal and industrial reuse demand is

sufficient to use reclaimed water at volumes that support cost recovery. By contrast, many water-scarce areas in northwestern, northeastern and southwestern China are characterized by dispersed settlements[53], weaker industrial demand[54] or limited profitable reuse[55] (Figs. 1c and 3d). In such settings, additional centralized treatment capacity can increase treatment coverage but generate limited marginal relief of local water scarcity. Approximately 420–500 million people live in regions where reclaimed water expansion can both recover its costs and alleviate local water scarcity (Fig. 3d). This figure represents the population living in regions where reuse meets both the cost-recovery and water-scarcity-mitigation criteria, rather than the number of people who would necessarily receive reclaimed water directly. Importantly, this spatial pattern remains stable across future socioeconomic pathways, indicating that projected urbanization and economic growth do not materially expand cost-recoverable reuse opportunities across most low-density, water-scarce regions.

An additional contribution is the empirically derived S-shaped relationship between collection-network density and wastewater collection rate (Fig. 3b). Large-scale wastewater assessments often represent collection rates as a fixed value or a linear response to sewer expansion[29]. Our results indicate that wastewater collection rate rises slowly while sewer networks remain sparse, increases rapidly after local connectivity becomes sufficient to link households and wastewater sources to trunk systems, and then approaches saturation as networks mature. This nonlinear relationship provides a way to estimate collection access from observable sewer density in regional and national wastewater models[23,25,30]. It can improve the representation of collection dynamics in studies that seek to link infrastructure investment with sludge generation[25], treatment coverage[56] or environmental exposure[10]. The numerical parameters of this relationship should be recalibrated for local topography[29], settlement morphology[35], road networks[57] and construction practices[58], but the underlying functional form offers a more realistic alternative to fixed coverage assumptions.

Decentralized wastewater treatment facilities enhance economic sustainability, reducing total investment costs by 25–37% relative to a CWTS-only pathway (Fig. 5a). Their advantage mainly reflects the reduced need for long-distance sewer conveyance (Fig. 4e), which avoids the steep increase in sewer costs associated with dispersed settlements[25,33,37]. By contrast, CWTSs require large service areas and long-lived, lumpy investments, making them more vulnerable to population redistribution, uncertain wastewater generation and low-load operation[59,60]. Decentralized systems serve smaller local catchments and can be deployed in phased, modular configurations, thereby reducing the risk of overbuilding and allowing treatment capacity and process complexity to be adjusted as local demand changes[37,61]. In rural and periurban settings with limited industrial contaminant inputs, nature-based or low-energy treatment units can support controlled near-source reuse, including agricultural reuse where effluent-quality and health-safety standards are met[12]. Thus, decentralized facilities are best deployed as a strategic complement to CWTSs where population density is low, sewer extension is costly, population trajectories are uncertain or near-source reuse opportunities are strong.

Economic assessment across the wastewater treatment lifecycle involves multiple stages, each of which relies on data subject to uncertainty. These uncertainties can propagate through the calculation process and affect the precision and robustness of the resulting estimates. National modelling of sewer collection remains uncertain, particularly because topography, settlement morphology, land-use dynamics and explicit spatial variation in labour and material costs are not fully represented. Technology selection and spatial layout are also simplified in the current framework, which does not explicitly account for energy and carbon constraints. In addition, although the optimization represents annual investment, asset retirement and reconstruction, it simplifies construction lead times, financing schedules, technology-specific renewal cycles and the possibility that many assets require replacement simultaneously.

These limitations do not alter the central planning implication of our analysis: centralized wastewater reuse is most effective where wastewater supply, reuse demand, water stress and financially viable service scales coincide, whereas hybrid systems offer a lower-cost and more resilient pathway where wastewater sources are dispersed and stable reuse demand is insufficient to support cost recovery. Future assessments could incorporate these additional spatial, technological and temporal dimensions to improve the realism of national wastewater planning and help design treatment systems that are not only technically feasible but also resilient, equitable and financially sustainable under rapid urbanization and climate uncertainty.

Methods

Data sources

To simulate water withdrawal and wastewater generation more precisely, this study integrated data from the Chinese Industrial Enterprise Database and the Chinese Industrial Enterprise Pollution Emission Database. Both databases were sourced from the China Microeconomic Data Query System (http://microdata.sozdata.com/). These two databases were utilized to pinpoint the specific locations, industrial sectors, water withdrawal and wastewater discharge of each enterprise (Supplementary Note 3 and Supplementary Figs. 20 and 21). On the basis of the total urban and rural domestic water withdrawal reported in China's Provincial Water Resources Bulletins, along with the urban and rural population from the county-level census dataset[62] (last accessed 1 September 2026) and land-use types (https://www.resdc.cn/), this study extracted higher-resolution grids for rural and urban domestic water withdrawal. For domestic point sources, wastewater generation was calculated by subtracting water consumption from water withdrawal. The water consumption rates were derived from China's Provincial Water Resources Bulletins.

The CWTS inventory was used to determine the spatial distribution and design capacity of existing treatment infrastructure. The CWTS inventory was compiled from multiple sources (Supplementary Fig. 2), including the annual statistical directories of WWTPs published by the Ministry of Ecology and Environment[63], the China Urban Drainage Statistical Yearbook (covering 2007–2018)[64] and the China Urban-Rural Construction Statistical Yearbook[65]. Sewer-network data were used to estimate the accessibility of wastewater sources to centralized treatment. Data on the collection network were sourced from the China Urban Construction Statistical Yearbook[66], the China County Construction Statistical Yearbook[67] and the China Urban-Rural Construction Statistical Yearbook[65]. Project-level investment data were used to derive scale-dependent infrastructure costs rather than applying a single national unit-cost assumption. WWTP fixed-asset investment and the capital cost of connection pipelines were compiled from 988 sewage-treatment public–private partnership (PPP) projects listed on China's National Development and Reform Commission PPP service platform (www.chinappp.cn). Connection pipelines denote the main trunk lines linking plants to collection areas, whereas the collection network comprises distributed local sewers.

Agricultural water withdrawal for cropping during 1980–2020 was estimated using the CROPWAT model[68] driven by effective rainfall and crop evapotranspiration from the China Surface Climate Data Daily Value Dataset (version 3.0) (http://data.cma.cn). Provincial natural water availability for 1980–2020 was downscaled to 0.25° × 0.25° on the basis of variable infiltration capacity model simulations from previous studies[69–71]. For future scenarios (2020–2050), grid-level projections of water availability and irrigation demand were obtained from ISIMIP3b, and future population projections were taken from Jing et al. and Huang et al.[53,72]. Overall, we used 1980–2020 as the historical baseline and simulated future scenarios for 2020–2050.

Assessment of water scarcity considering wastewater treatment

WSI was used to quantify whether reclaimed water use actually reduces pressure on local freshwater resources. WSI compares freshwater demand with locally available water resources and therefore allows the water-security benefit of wastewater reuse to be evaluated consistently across regions[2,73]. The WSI is measured as the ratio of regional water withdrawal to available water resources

$$\mathrm{WSI} = \frac{\sum_i D_i - \mathrm{RW}_i}{Q - \mathrm{EFR}}, \tag{1}$$

where D_i represents the water withdrawal by sector i, Q is the total water resource, and EFR denotes the environmental flow requirement, defined as 80% of the total water resource, following previous studies[2,73]. RW_i refers to the amount of reclaimed water reused by sector i. Although treated wastewater meeting discharge standards has been considered by some studies as a potential source for augmenting surface water availability[14], from the perspective of a strict definition of water scarcity, only reclaimed water that actually reduces freshwater withdrawals can contribute to alleviating water stress. Following the settings of previous research[2,11], WSI values were classified into seven water scarcity levels: low (0–0.5), moderate (0.5–1.0), high (1.0–1.5), very high (1.5–2.0), severe (2.0–4.0), extreme (4.0–6.0) and very extreme (>6.0). The water scarcity benefit of reuse was expressed as the percentage reduction in WSI relative to the corresponding no-reuse condition, denoted ΔWSI.

Material flow analysis

The material flow analysis tracks water from sectoral withdrawal to wastewater generation, sewer collection and potential reuse. This step was needed because total water withdrawal cannot be treated as wastewater supply: part of the withdrawn water is consumed, and part of the resulting wastewater is generated outside areas accessible to centralized sewers. Using the Chinese Industrial Enterprise Database, the Chinese Environmental Database and the Water Resources Bulletin (Supplementary Notes 1 and 2 and Supplementary Tables 6–8), we estimated grid-level water withdrawal and wastewater discharge. Water-withdrawal estimates cover all sectors, whereas wastewater discharge estimates include only collectable industrial and domestic point sources. The calculation formulas are as follows:

$$\begin{aligned}\mathrm{WU}_{y,i} = {} & \sum_j \mathrm{IWU}_{\mathrm{irr},y,i,j} \times A_{\mathrm{irr},y,i,j} + \sum_j \mathrm{IWU}_{\mathrm{ind},y,i,j} \times \mathrm{GDP}_{\mathrm{ind},y,i,j} \\ & + \sum_j \mathrm{IWU}_{\mathrm{urb},y,i} \times \mathrm{POP}_{\mathrm{urb},y,i} + \sum_j \mathrm{IWU}_{\mathrm{rur},y,i} \times \mathrm{POP}_{\mathrm{rur},y,i}\end{aligned} \tag{2}$$

$$\begin{aligned}\mathrm{WW}_{y,i} = {} & \mathrm{WU}_{\mathrm{ind},y,i,j} \times (1 - C_{\mathrm{ind},y,i,j}) + \mathrm{WU}_{\mathrm{urb},y,i} \\ & \times (1 - C_{\mathrm{urb},y,i}) + \mathrm{WU}_{\mathrm{rur},y,i} \times (1 - C_{\mathrm{rur},y,i}),\end{aligned} \tag{3}$$

where y denotes the year, i represents the grid cell index and j represents the crop or industry type. WU, WW, IWU, A, GDP, POP and C represent total water withdrawal, total wastewater generation, water withdrawal intensity, irrigated area, gross domestic product, population and the water consumption rate, respectively. Variables irr, ind, urb and rur correspond to water withdrawals for irrigation, industry, urban domestic, and rural domestic use, respectively.

Because wastewater generation does not necessarily translate into wastewater that can be collected by centralized sewer networks, we defined collectable wastewater according to infrastructure service demand rather than total water-use location. This definition is consistent with China's land-use planning system[74], in which land use is regulated by planned functional purposes; dense sewer networks are therefore most realistically deployed in residential and industrial construction areas rather than across all land-use types[75]. Domestic collectable wastewater was allocated to urban and rural residential land, where sewer infrastructure serves population settlements, whereas industrial collectable wastewater was constrained to industrial enterprise points. Agricultural water use and dispersed non-built-up domestic water use were not treated as directly recoverable wastewater supply for centralized systems, because much of this water is consumed, infiltrated or discharged outside sewer-accessible areas. This treatment is consistent with the higher consumption rates assigned to rural domestic water use, reflecting more dispersed wastewater discharge and lower sewer connectivity (Supplementary Fig. 13). Thus, residential land and industrial land were used as spatial constraints for infrastructure service objects, not as assumptions that all water use within these land types generates sewer-accessible wastewater. Robustness checks using land-use grids from different years and comparisons with national field-survey land-use data support the consistency of this spatial boundary (Supplementary Fig. 22).

Available national datasets provide WWTP locations and design capacities but do not consistently report plant-level sewer catchment boundaries. We therefore reconstructed approximate service areas for individual WWTPs. On the basis of the linear relationship between the collection area and design capacity, the collection area for every WWTP was simulated, centred at the WWTP location. As large-scale inter-prefecture wastewater collection is currently uncommon in China[26], the collection areas for WWTPs located near prefectural boundaries were adjusted accordingly. WWTPs whose collection areas were fully contained within that of another WWTP were treated as a single WWTP. The collection-network density was used as a spatially explicit proxy for sewer accessibility because reported coverage rates do not describe how intensively sewers serve individual areas. The wastewater collection rate ($\mathrm{Col}_{y,i}$) was therefore estimated as a nonlinear function of local collection-network density ($\mathrm{pipe}_{y,i}$) (Supplementary Note 4 and Supplementary Fig. 6). The volume of wastewater treated ($\mathrm{WWT}_{y,m}$) was calculated using the formula detailed below.

$$\mathrm{WWT}_{y,m} = \sum \mathrm{WW}_{y,i} \times \mathrm{Col}_{y,i} \tag{4}$$

$$\mathrm{Col}_{y,i} = f(\mathrm{pipe}_{y,i}) \tag{5}$$

WWTP capacity adequacy was assessed by comparing design capacity (DC_m) with collected wastewater within each service area. This comparison separates areas where limited treatment is caused by insufficient plant capacity from those where capacity exists but wastewater cannot be effectively collected. Plant load rate was then used to classify operating conditions.

$$\text{Adequacy level} = \begin{cases} \text{Surplus capacity} & \mathrm{DC}_m \times 0.6 > \mathrm{WWT}_{y,m} \\ \text{Balanced capacity} & \mathrm{DC}_m \times 0.6 \le \mathrm{WWT}_{y,m} \le \mathrm{DC}_m \times 1.2 \\ \text{Insufficient capacity} & \mathrm{DC}_m \times 1.2 < \mathrm{WWT}_{y,m} \end{cases} \tag{6}$$

Reclaimed water utilization potential ($\mathrm{RP}_{n,t}$) was defined as the minimum of sectoral reuse demand and wastewater supply. Sectoral demand comprised irrigation ($D^{\mathrm{irr}}_{n,t}$), domestic non-potable use ($D^{\mathrm{dom}}_{n,t}$) and industrial reuse ($D^{\mathrm{ind}}_{n,t}$), whereas wastewater supply included domestic ($\mathrm{WW}^{\mathrm{dom}}_{n,t}$) and industrial wastewater generation ($\mathrm{WW}^{\mathrm{ind}}_{n,t}$). High-quality reclaimed water potential ($\mathrm{HRP}_{n,t}$) was restricted to non-potable residential use and industrial cooling use, because they were treated as profitable reuse categories (Supplementary Fig. 9). Based on previous studies, reclaimed water can substitute for about 30–70% of industrial cooling water[76] and 20–40% of non-potable domestic uses (mainly toilet flushing)[77]. In rural areas, low-quality treated wastewater can be directly reused for irrigation. To account for water scarcity, reclaimed water utilization potentials were further constrained by local water shortage. Water shortage at grid n in year t ($\mathrm{WS}_{n,t}$) was quantified from

the projected WSI as the excess above unity, scaled by local renewable water availability.

$$\mathrm{RP}_{n,t} = \min\left(D^{\mathrm{irr}}_{n,t} + D^{\mathrm{dom}}_{n,t} + D^{\mathrm{ind}}_{n,t}, \mathrm{WW}^{\mathrm{dom}}_{n,t} + \mathrm{WW}^{\mathrm{ind}}_{n,t}\right) \quad (7)$$

$$\mathrm{HRP}_{n,t} = \min\left(D^{\mathrm{dom}}_{n,t} + D^{\mathrm{ind}}_{n,t}, \mathrm{WW}^{\mathrm{dom}}_{n,t} + WW^{\mathrm{ind}}_{n,t}\right) \quad (8)$$

$$\mathrm{RP}^{\mathrm{scar}}_{n,t} = \min(D^{\mathrm{irr}}_{n,t} + D^{\mathrm{dom}}_{n,t} + D^{\mathrm{ind}}_{n,t}, \mathrm{WW}^{\mathrm{dom}}_{n,t} + \mathrm{WW}^{\mathrm{ind}}_{n,t}, \mathrm{WS}_{n,t}) \quad (9)$$

$$\mathrm{HRP}^{\mathrm{scar}}_{n,t} = \min(D^{\mathrm{dom}}_{n,t} + D^{\mathrm{ind}}_{n,t}, \mathrm{WW}^{\mathrm{dom}}_{n,t} + \mathrm{WW}^{\mathrm{ind}}_{n,t}, \mathrm{WS}_{n,t}) \quad (10)$$

To identify where reclaimed water reuse provides robust and policy-relevant benefits, we classified each grid cell according to two criteria across 1,000 Monte Carlo realizations that sampled uncertainty in socioeconomic pathway, projection year, wastewater generation and water-use structure. The first criterion tested whether locally available reclaimed water could reduce the WSI by more than 10% (ΔWSI >10%); the second tested whether economically favourable reclaimed water demand could absorb at least 60% of local wastewater generation, consistent with the national reclaimed water utilization target. A criterion was considered robust only when it was satisfied in at least 95% of the realizations. Grid cells were classified as showing substantial water scarcity mitigation when only the ΔWSI >10% criterion was robustly met, as meeting the cost-recoverable reuse target when only the 60% profitable reuse criterion was robustly met, and as satisfying both conditions when both criteria were robustly met. Cells were classified as not substantial when both criteria were unmet in at least 95% of the realizations, whereas cells without any stable 95% classification were labelled as having no robust condition.

Calibration of economic viability thresholds

The next question is when wastewater treatment and reclaimed water production can recover their costs. We therefore estimated two complementary thresholds: the minimum treatment scale required for wastewater treatment cost recovery and the reclaimed water end-use mix required for reuse revenues to cover production costs. For treatment-scale thresholds, unit treatment costs were estimated using the two-level Monte Carlo framework (Supplementary Note 7 and Supplementary Figs. 24–26). Existing WWTPs were evaluated individually, whereas currently unserved areas were evaluated using 5 km × 5 km grid cells as potential service units. This grid size was selected as a conservative representation of the lower-range service-area scale implied by the empirical relationship between WWTP design capacity and catchment area (Supplementary Fig. 11). Observations were grouped into 10 m^3 per day design-capacity intervals, and the minimum and maximum unit costs in each interval were retained to construct empirical lower- and upper-cost frontiers, respectively (Supplementary Fig. 27). A tariff below the lower-cost frontier cannot recover costs even under the most favourable observed cost conditions, whereas a tariff above the upper-cost frontier can recover costs across the observed cost range; tariffs between the two frontiers are condition dependent. Restricted cubic splines were used to smooth this frontier across plant scales. Repeated Monte Carlo realizations generated uncertainty bands, with the 5th, 50th and 95th percentiles representing optimistic, median and conservative cost boundaries. Economically viable treatment-scale thresholds were defined as the intersections between these lower-cost frontiers and alternative wastewater treatment tariffs.

For reclaimed water production, we compiled 141 project cases and estimated the end-use composition required for cost recovery using a Monte Carlo portfolio analysis. Reclaimed water uses were grouped into groundwater recharge, landscape use, non-potable residential use, industrial process water, industrial cooling water and irrigation. We compiled empirical unit-cost samples for each end-use category and combined them with the corresponding reclaimed water revenues. For each tested share of profitable reuse, ranging from 0 to 100%, reclaimed water allocation within profitable reuse and remaining reuse was randomly generated using Dirichlet sampling. Portfolio revenue and cost were calculated as demand-weighted averages across end uses, and cost recovery was defined as revenue exceeding cost. The cost-recovery threshold for reclaimed water was defined as the minimum share of profitable reuse required to reach a 95% cost-recovery probability across 1,000 Monte Carlo realizations. This threshold identifies the end-use composition at which cost recovery is robust to uncertainty rather than achievable only under favourable parameter combinations. Economically favourable end uses were defined as categories whose median empirical unit cost was lower than the median reclaimed water tariff. On the basis of this criterion, non-potable residential use and industrial cooling water were treated as profitable reuse categories in the main portfolio analysis (Supplementary Fig. 9).

Cost quantification for wastewater treatment solutions

In China, the wastewater treatment systems are mainly divided into two categories: CWTSs and decentralized wastewater treatment systems (DWTSs)[19,47,61,78,79]. Costs for CWTSs include: construction costs (for collection networks, long-distance connection pipelines and WWTPs), upgrading costs (for discharge standards), production costs (for reclaimed water), land acquisition costs and operating costs[80,81] (Fig. 4a, Supplementary Fig. 23 and Supplementary Table 5). The costs of the decentralized systems include the construction of sewage treatment facilities, operating costs, the construction of collection networks and land acquisition costs[82–89] (Supplementary Table 5).

To estimate CWTS fixed-asset costs, we compiled 988 sewage-treatment PPP projects from the National Development and Reform Commission PPP platform. After excluding projects without key technical information, 766 projects were retained. Because the database reports total investment rather than component-level expenditure, we first used multivariable regression to identify the relative contributions of WWTP capacity, pipeline length, upgrade scale, reclaimed water capacity and urban–rural location to project investment (Supplementary Note 5 and Supplementary Tables 9 and 10). Projects were then grouped by their primary investment object (WWTP construction, connection pipelines, treatment upgrading or reclaimed water production), and unit-cost functions were estimated separately for each group (Supplementary Fig. 12). Operating costs were obtained from the Urban Drainage Statistical Yearbook[64]. The total cost of CWTSs was calculated using the following formula:

$$\mathrm{ACT}_m = \mathrm{ACF}_m + \mathrm{ACO}_m \quad (11)$$

$$\mathrm{ACF}_m = \mathrm{CF}_m/\mathrm{YS} = \left(\begin{array}{l}\mathrm{SD}_m \times \mathrm{CWCost}_m + \mathrm{Pipe}_m \times \mathrm{PCost}_m + \mathit{C}\mathrm{Pipe}_m \times \mathit{P}\mathrm{Cost}_m \\ +\mathrm{SD}_m \times \mathrm{CUCost}_m + \mathrm{SD}_m \times \mathrm{CRCost}_m + \mathrm{Land}_m \times \mathrm{CLCost}_m\end{array}\right)/\mathrm{YS} \quad (12)$$

$$\mathrm{ACO}_{m,y} = \left(\mathrm{WWT}_{m,y} \times \mathrm{COCost}_m\right) \quad (13)$$

ACT_m, ACF_m, CF_m and $\mathrm{ACO}_{m,y}$ denote the average investment per year, annual depreciation cost, total fixed-asset investment and annual operating cost of CWTS m, respectively. SD_m, Pipe_m, $\mathit{C}\mathrm{Pipe}_m$ and Land_m represent the design capacity, collection pipe length, connection pipe length and land footprint of CWTS m, respectively. CWCost_m, $\mathit{P}\mathrm{Cost}_m$, CUCost_m, CRCost_m, CLCost_m and COCost_m correspond to the unit costs of WWTPs, pipelines, effluent standards enhancement, reclaimed water production, land acquisition and operational activities of CWTS m, respectively. YS refers to the design service life, which is typically standardized as 20–40 years for infrastructure planning[39]. Cost calculations for DWTSs were based on the Guideline on Project Construction

and Investment for Rural Sewage Treatment[79], with computational formulas specified as follows:

$$\mathrm{ADT}_m = \mathrm{ADF}_m + \mathrm{ADO}_m \tag{14}$$

$$\mathrm{ADF}_m = \mathrm{DF}_m/\mathrm{YS} = \left(\mathrm{SD}_m \times \mathrm{DWCost}_m + \mathrm{Pipe}_m \times P\mathrm{Cost}_m + \mathrm{Land}_m \times \mathrm{DLCost}_m\right)/YS \tag{15}$$

$$\mathrm{ADO}_{m,y} = \left(\mathrm{WWT}_{m,y} \times \mathrm{DOCost}_m\right) \tag{16}$$

ADT_m, ADF_m, DF_m and $\mathrm{ADO}_{m,y}$ denote the average investment per year, average annual depreciation cost, total fixed-asset investment and annual operational investment of decentralized sewage treatment systems m, respectively. DWCost_m, DLCost_m and DOCost_m correspond to the unit costs of sewage treatment facilities, land acquisition and operational activities of decentralized sewage treatment systems m. As the decentralized facilities handle a relatively small volume of sewage, the cost of additional treatment to produce reclaimed water was not considered. The design service life (YS) varies with the type of DWTS selected[82–89] (Supplementary Table 13).

Optimization model for wastewater infrastructure

The optimization determines which infrastructure action should be implemented in each service unit and year. Its objective is to meet national treatment and reuse targets while maximizing long-term net benefit. We developed an integrated optimization framework to identify the most cost-effective wastewater management pathways across coupled climate and socioeconomic futures, represented by alternative Representative Concentration Pathway (RCP)–SSP combinations (Supplementary Fig. 28). Climate scenarios influence wastewater generation through changes in water availability, while socioeconomic pathways determine population dynamics, urbanization patterns and sectoral water demand, jointly shaping the evolution of wastewater flows, treatment demand and reclamation potential.

The model was designed around explicit policy targets to be achieved by 2035, including 100% urban wastewater treatment, 85% rural wastewater treatment and 60% reclamation rate of treated effluent. These targets were imposed as binding constraints. Optimization was conducted over the period 2020–2050, allowing the system to adapt dynamically to infrastructure ageing, CWTS retirement waves and population redistribution beyond the target year. The post-2035 simulation primarily served to assess the long-term cost implications of maintaining policy targets under changing demographic and climatic conditions. We also applied a scenario that relies exclusively on CWTSs for comparison.

The objective function maximized the total net benefit of the national wastewater management system by 2050. For each candidate scheme m in year y, net benefit was defined as:

$$\mathrm{NBef}_{m,y} = \mathrm{Rev}_{m,y} - \mathrm{Depr}_{m,y} - \mathrm{Opex}_{m,y} \tag{17}$$

$$\mathrm{TNBef}_m = \sum_y \mathrm{NBef}_{m,y}, \tag{18}$$

where TNBef_m and $\mathrm{NBef}_{m,y}$ denote the total net benefit and annual net benefit of scheme m, respectively, $\mathrm{Rev}_{m,y}$ represents the annual revenue, $\mathrm{Depr}_{m,y}$ refers to the depreciation cost of fixed assets, and $\mathrm{Opex}_{m,y}$ indicates the operating costs. Capital investments were incorporated through asset depreciation, ensuring consistency between short-term implementation decisions and long-term financial performance.

Two investment-pacing strategies were compared to test whether the timing of infrastructure deployment affects long-term economic performance. At each annual time step, candidate schemes were ranked according to their marginal net benefit. The implementation followed one of two alternative progression strategies. Under the IC strategy, annual expenditures were constrained by a predefined budget ceiling derived from the aggregate cost of all feasible schemes, and implementation proceeded until the budget was exhausted. Under the GO strategy, annual investment was not subject to a fixed budget cap. Instead, the model required investments sufficient to close one-fifteenth of the remaining gap between the current system state and the 2035 management targets each year, thereby ensuring that the targets were achieved by 2035. This pacing reflected typical medium-term infrastructure planning cycles and avoided front-loading of low-yield investments, but it could lead to excessive investment in a single year.

Existing service areas were allowed to modify or consolidate current assets, whereas unserved areas could either build new centralized infrastructure or adopt decentralized alternatives. For service areas (regions already served by CWTS), five optimization options were available in addition to decommissioning and reconstruction: (1) premature decommissioning and diversion to nearby treatment plants, (2) pipeline network expansion, (3) capacity expansion, (4) implementation of wastewater reclamation and (5) downsizing of over-designed capacity. In unserved regions, six alternatives were considered: (1) construction of new WWTPs, (2) collection and transfer to nearby facilities, (3) surface-flow-constructed wetlands, (4) subsurface-flow-constructed wetlands, (5) stabilization ponds and (6) integrated treatment units. To ensure realistic implementation, only one scheme could be applied in a given region within a single year.

Each implemented scheme updated two coupled stock-and-flow ledgers. The financial ledger recorded residual fixed assets, operating costs, wastewater treatment revenue, reclaimed water revenue and marginal financial changes induced by new investments. The wastewater ledger tracked collected volume, treated volume, overload volume, reclaimed water production and corresponding marginal changes. The optimization was further constrained by agricultural land availability for irrigation reuse and industrial and domestic demand for reclaimed water (Supplementary Note 5). Under the IC strategy, implementation was additionally limited by a fixed annual investment ceiling.

Climate and socioeconomic scenarios

The integrated modelling framework was implemented under two combined climate–socioeconomic scenarios, pairing RCPs with SSPs for the period up to 2050: RCP2.6–SSP1 and RCP8.5–SSP5. These two combinations were selected to represent contrasting yet policy-relevant development pathways for China's wastewater sector. The RCP2.6–SSP1 scenario reflects a sustainability-oriented pathway characterized by low radiative forcing, moderated urbanization and resource-efficient economic growth, whereas RCP8.5–SSP5 represents a fossil-fuelled development trajectory with high greenhouse gas emissions, rapid urbanization and strong economic growth, resulting in substantially higher pressure on water resources and wastewater infrastructure.

Urban population, rural population and industrial output projections for 2050 under SSP1 and SSP5 were adopted as the primary socioeconomic drivers, based on Huang et al.[72], and were used to construct annual population and urbanization trajectories over the simulation period. Future industrial water withdrawal and wastewater generation were estimated by coupling data-driven intensity forecasting with scenario-based industrial output projections. Specifically, the exponential decay method was used to extrapolate historical trends in water use intensity and wastewater generation intensity per unit of industrial output, capturing both long-term structural changes and nonlinear temporal dependencies. Model performance was evaluated using out-of-sample validation, and details of model structure, training and error metrics are provided in Supplementary Figs. 29 and 30.

For future optimization, industrial pretreatment rates were varied by ±20% around the 2020 baseline as a conservative sensitivity interval[90], representing plausible variation in future pretreatment compliance and enforcement intensity without introducing an additional regulatory-policy scenario.

Statistics and reproducibility

No statistical method was used to predetermine sample size. Sample sizes were determined by the availability and completeness of the national inventories and project datasets used in each analysis. The national WWTP inventory comprised 10,826 plants. For estimation of CWTS fixed-asset costs, 988 PPP projects were compiled; projects lacking key technical information were excluded, leaving 766 projects for regression and cost calibration. The reclaimed water cost analysis included 141 project cases. Uncertainty in socioeconomic pathways, wastewater generation, water-use structure and cost parameters was propagated using Monte Carlo simulations. Unless otherwise stated, Monte Carlo results are summarized by the median (50th percentile) and the 5th–95th percentile interval from 1,000 realizations, and robust classifications required the relevant criterion to be satisfied in at least 95% of realizations. The experiments were not randomized; random sampling was used only within computational Monte Carlo and Dirichlet-sampling procedures. The investigators were not blinded to allocation during analyses and outcome assessment because the study used observational datasets and predefined computational models rather than experimental interventions. Exact sample sizes, definitions of central estimates and uncertainty intervals, and analysis-specific statistical procedures are provided in the corresponding Methods subsections and figure legends.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper. Additional data generated or compiled in this study, including the WWTP inventory (plant locations, design capacities and annual treatment volumes), reconstructed WWTP service-area data, reclaimed water project data and PPP wastewater treatment investment project data, are provided as Supplementary Data.

Code availability

Data collection and analyses were conducted in Microsoft Excel 2020, Python version 3.11.9, ArcGIS version 10.8.2 and Origin 2024b. The main code created in this work is available via Zenodo at https://doi.org/10.5281/zenodo.22138940 (ref. 91).

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Author contributions

J.S., L.S., S.Z. and Delong L. designed the research; J.S., L.S., S.Z. and Delong L. performed the research; J.S., Ding L., J.F. and H.B. collected and analysed the data; J.S., L.S., S.Z. and Delong L. wrote the Article; and L.S., S.Z. and Delong L. reviewed and edited the manuscript.

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Competing interests

The authors declare no competing interests.

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Data collection	No software was used for data collection.
Data analysis	Data analyses were conducted in Microsoft Excel 2020, Python v.3.11.9, ArcGIS v.10.8.2 and Origin 2024b. Custom Python code used for the analyses reported in this study has been deposited in Zenodo at https://doi.org/10.5281/zenodo.22138940.

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Study description	This is a quantitative observational and computational modelling study of China's centralized wastewater treatment system. We integrated national wastewater-treatment-plant inventories with spatially explicit wastewater generation, sewer-network, water-use, reclaimed-water-demand, tariff and project-cost datasets to evaluate the spatial and economic limits of centralized wastewater treatment and reuse. Historical patterns were reconstructed and future centralized-only and hybrid centralized–decentralized pathways were evaluated under contrasting climate–socioeconomic scenarios through 2050 using uncertainty analysis and dynamic infrastructure optimization.
Research sample	The study used existing national inventories, project datasets and spatial datasets rather than biological or experimental samples. The main empirical units included a national inventory of 10,826 wastewater treatment plants, 988 wastewater-treatment PPP projects (766 retained for fixed-asset cost analysis after data-quality screening), and 141 reclaimed-water project cases, together with national socioeconomic, water-use, sewer-network and gridded environmental datasets. These datasets were selected to provide national-scale coverage of wastewater generation, collection, treatment, reuse demand and infrastructure costs in China.
Sampling strategy	No statistical method was used to predetermine sample size. Sample sizes were determined by the availability and completeness of the national inventories and project datasets required for each analysis. The WWTP inventory was compiled to achieve national coverage rather than through statistical sampling. PPP and reclaimed-water project records were retained when the information required for the corresponding analysis was available. Analysis-specific sample sizes are reported in the Methods, figure legends and Supplementary Information.
Data collection	The inventory of centralized wastewater treatment plants was compiled from multiple sources, primarily utilizing the annual statistical directories published by the Ministry of Ecology and Environment for the years 2007 to 2014 and 2020.
Timing and spatial scale	Timing scale: 1980-2050 Spatial scale: Precise locations in China mainland
Data exclusions	Data exclusions were based on data-quality and completeness criteria rather than study outcomes. For the PPP dataset, 988 projects were compiled and projects lacking key technical information required for cost analysis were excluded, leaving 766 projects. Additional cleaning of the industrial-enterprise datasets addressed missing or erroneous records as described in Supplementary Note 3. No experimental subjects or observations were excluded on the basis of the model outcomes.
Reproducibility	The results can be reproduced.
Randomization	Randomization is no relevant to our study.
Blinding	Blinding is not relevant to this study, as no treatment is applied.

Did the study involve field work? ☐ Yes ☒ No

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Plants

Seed stocks	N/A
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Authentication	N/A